Experiment – 9

Android Calculator Application using Android Studio

AIM:

*To create an Android application that performs basic arithmetic operations using Android UI components such as **Button, TextView, and EditText**. The application should support operations like **Addition, Subtraction, Multiplication, and Division**.*

ALGORITHM:

1. ***Create the Android Project:***
   - *Open Android Studio and start a new project.*
   - *Create the main activity file (Java/Kotlin) for the application interface.*
2. ***Design the Layout:***
   - *Build the user interface using XML.*
   - *Add the following components:*
     - *Two **EditText** fields for entering numbers.*
     - *Four **Buttons** for operations: Add, Subtract, Multiply, Divide.*
     - *One **TextView** to show the result.*
3. ***Initialize Variables:***
   - *Declare variables for all UI elements (EditText, Buttons, TextView).*
   - *Create variables to store input values and result (e.g., num1, num2, result).*

4. ***Read User Input:***

- *Retrieve input values from EditText fields using getText().toString().*
- *Convert the input strings into numeric values using methods like Integer.parseInt() or Double.parseDouble().*

5. ***Add Button Click Events:***
   - *Attach OnClickListener to each button.*
   - *Define actions inside the onClick() method for each operation.*
6. ***Execute Operations:***
   - *Based on the button clicked, perform the corresponding calculation:*
     - *Addition → result = num1 + num2*
     - *Subtraction → result = num1 - num2*
     - *Multiplication → result = num1 * num2*
     - *Division → result = num1 / num2*
7. ***Handle Division Error:***
   - *Before division, check if the second number is zero.*
   - *If true, display an error message instead of performing the operation.*
8. ***Display Output:***
   - *Convert the result into a string format.*
   - *Show the result in the TextView using setText().*
9. ***Validate Inputs:***
   - *Ensure both input fields are not empty.*
   - *Check that valid numeric values are entered before performing calculations.*

10. ***Completion:***

- *Display the calculated result to the user.*
- *Allow the user to perform multiple calculations or exit the application.*

CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Simple Calculator"
        android:textStyle="bold"
        android:textSize="30sp"
        android:layout_gravity="center"
        android:layout_marginTop="50dp"/>

    <EditText
        android:id="@+id/number1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter number 1"
```

```
    android:inputType="number"
    android:layout_marginTop="50dp"/>

<EditText
    android:id="@+id/number2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter number 2"
    android:inputType="number"
    android:layout_marginTop="20dp"/>

<LinearLayout
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginTop="30dp">

    <Button
        android:id="@+id/btn_add"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="+"
        android:textSize="30sp"/>

    <Button
        android:id="@+id/btn_sub"
        android:layout_width="wrap_content"
```

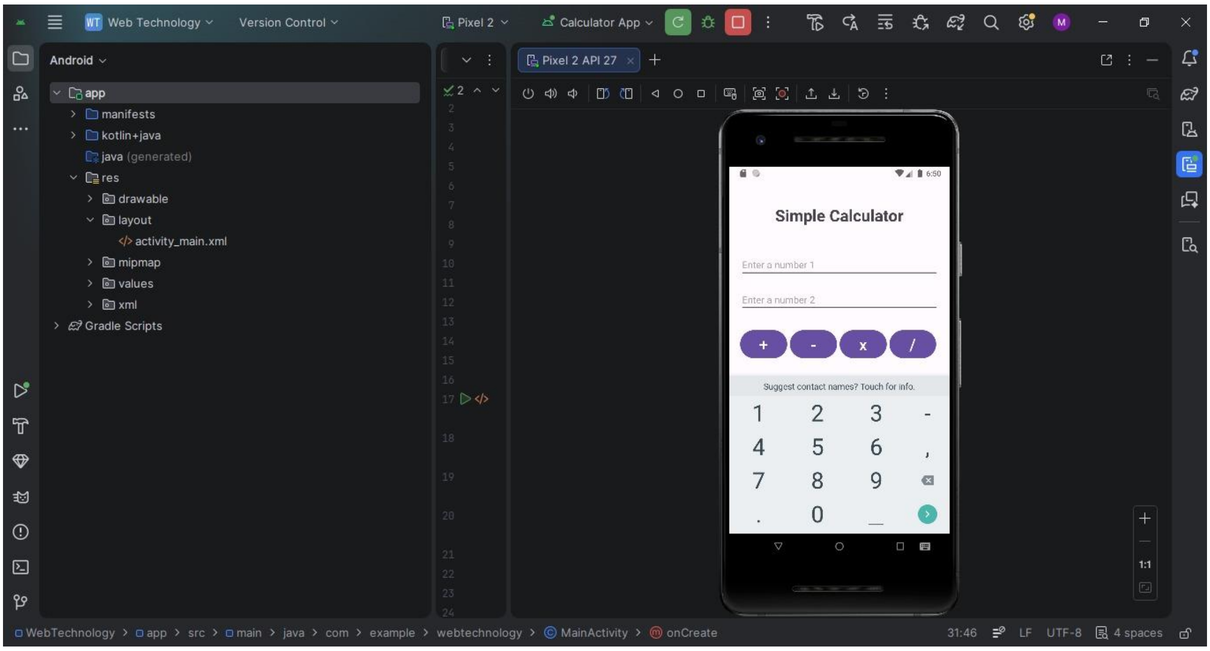
```
        android:layout_height="wrap_content"
        android:text="-"
        android:textSize="30sp"
        android:layout_marginStart="5dp"/>

    <Button
        android:id="@+id/btn_mul"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="x"
        android:textSize="30sp"
        android:layout_marginStart="5dp"/>

    <Button
        android:id="@+id/btn_div"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="/"
        android:textSize="30sp"
        android:layout_marginStart="5dp"/>

</LinearLayout>

<TextView
    android:id="@+id/answer"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
        android:textSize="30sp"
        android:textStyle="bold"
        android:layout_marginTop="30dp"
        android:layout_gravity="center"/>

</LinearLayout>
package com.example.webtechnology;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity implements
View.OnClickListener {

    Button buttonAdd, buttonSub, buttonMul, buttonDiv;
    EditText editTextn1, editTextn2;
    TextView textview;
    int num1, num2;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

        buttonAdd = findViewById(R.id.btn_add);
        buttonSub = findViewById(R.id.btn_sub);
        buttonMul = findViewById(R.id.btn_mul);
        buttonDiv = findViewById(R.id.btn_div);

        editTextn1 = findViewById(R.id.number1);
        editTextn2 = findViewById(R.id.number2);

        textview = findViewById(R.id.answer);

        buttonAdd.setOnClickListener(this);
        buttonSub.setOnClickListener(this);
        buttonMul.setOnClickListener(this);
        buttonDiv.setOnClickListener(this);
    }

    public int getIntFromEditText(EditText editText) {
        if (editText.getText().toString().trim().equals("")) {
            Toast.makeText(this, "Enter number",
Toast.LENGTH_SHORT).show();
            return 0;
        } else {
            return Integer.parseInt(editText.getText().toString());
        }
    }
```

```
    @Override
    public void onClick(View v) {

        num1 = getIntFromEditText(editTextn1);
        num2 = getIntFromEditText(editTextn2);

        int id = v.getId();

        if (id == R.id.btn_add) {
            textview.setText("Answer = " + (num1 + num2));

        } else if (id == R.id.btn_sub) {
            textview.setText("Answer = " + (num1 - num2));

        } else if (id == R.id.btn_mul) {
            textview.setText("Answer = " + (num1 * num2));

        } else if (id == R.id.btn_div) {

            if (num2 == 0) {
                Toast.makeText(this, "Cannot divide by zero",
Toast.LENGTH_SHORT).show();
            } else {
                textview.setText("Answer = " + ((float) num1 / num2));
            }
        }
    }
}
```

OUTPUT:



RESULT:

A Calculator is designed using android studio which uses controls like button,textview and performs functions like addition,subtraction,multiplication and division.